Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

Francisco Hernando-Gallego,*,† Dimitar Vasilev-Vasilev,‡ Diego García-Santos,‡ Alfonso Campo-Camacho,¶ and Alberto Gómez-Gómez¶

†*Department of Applied Mathematics, Escuela de Ingeniería Informática de Segovia, Universidad de Valladolid, 40005 Segovia, Spain*

‡*Department of Computer Science, Escuela de Ingeniería Informática de Segovia, Universidad de Valladolid, 40005 Segovia, Spain*

¶*Independent Researcher, Spain*

E-mail: fhernando@uva.es

Abstract

High-throughput molecular dynamics (MD) produces the labels on which electrolyte screening and machine-learning models are built, but each label comes from one short, noisy run whose value still depends on the simulated time. We quantify their reliability, and how to spend a simulation budget given it, on HTP-MD, the largest public set of polymer electrolyte trajectories (6,137 runs of 5 ns and 133 of 50 ns). Lithium motion is diffusive in only 2 short and 5 long runs, the apparent diffusivity at a 5 ns lag is 1.95 times that at 25 ns, and the first 5 ns of a run are 1.63 times faster than its second half, biasing every short label upwards. From 94 systems simulated twice, a single

5 ns end-point label has reliability 0.47 (95% CI 0.31 to 0.63): no model can correlate with such labels much beyond about 0.69, a random forest on monomer fingerprints is indistinguishable from that ceiling (0.71), and roughly half of the run-to-run noise comes from the initial structure. A short run ranks widely different polymers well (Spearman 0.93) but not similar ones (0.68 within the screening set). In simulations with the measured noise, giving further runs only to candidates whose membership of the top set is uncertain matches four uniform replicates with 1.8 runs per candidate for the top 10% and 1.2 for the top 1%; a check on real runs is consistent but underpowered. A decision guide and all code are provided.

Introduction

Molecular dynamics (MD) is increasingly used not to study one material carefully but to label thousands of candidates, either to rank them directly or to train machine-learning models that extend the search.[1] Solid polymer electrolytes are a representative case: lithium transport is slaved to the slow segmental dynamics of the polymer,[2,3] so the simulations needed to converge transport coefficients are long, while the chemical space is large. The HTP-MD platform, the largest public collection of such simulations, distributes thousands of short runs of 5 ns and a smaller number of 50 ns runs, with raw trajectories and derived properties.[1,4]

A label obtained this way has two defects that are easy to overlook once it sits in a table. It is *time dependent*: the self-diffusion coefficient is defined in the long-time limit, a diffusivity extracted before the mean-squared displacement (MSD) becomes linear is a property of the observation window,[5] and the window of a screening run is short. It is also *noisy*: the amorphous structure is sampled once, and two runs of the same polymer from different initial structures can disagree substantially.[1] The dataset's authors addressed both defects by training a multi-task graph network on many short and few long runs, assuming noise independent of the polymer.[1] Learning from noisy labels is well studied in

machine learning,[6] and the statistics of diffusion estimates from finite trajectories are well understood.[7,8,9] That measurement noise bounds the performance any model can show is also established: for experimental activity data, the replicate variability sets the maximum correlation or R^2 a model can reach,[10,11] and computational neuroscience estimates the same bound as a noise ceiling.[12] For simulated labels, however, to our knowledge the noise across a real screening campaign has not been measured, although the same question has been raised for chemistry datasets in general;[13] its sources have not been separated, whether a model actually reaches the resulting ceiling has not been tested, and the noise has not been used to decide how a fixed simulation budget should be spent. Unlike experimental noise, simulation noise can be reduced by spending more computation, which turns the ceiling into a question of allocation.

We answer these questions on HTP-MD without training a corrective model, using three features of the public collection: 94 screening systems simulated twice from different initial structures, 133 systems simulated at both lengths, and the raw long trajectories, which can be truncated to emulate shorter runs. Our contributions are:

1. **Time dependence of the labels.** A census of all 6,270 trajectories, the lag dependence of the apparent diffusivity in the long runs, and a slow decrease of lithium mobility within each run that makes its beginning faster than its later part.

2. **Reliability and noise ceiling.** The run-to-run noise of a short label, its structural and dynamic sources and its dependence on mobility, the ceiling it imposes on any predictor trained or evaluated on such labels, a machine-learning model that reaches that ceiling, and the ability of one short run to select the best candidates, checked on a second property, the anion diffusivity.

3. **Spending the budget.** An evaluation of estimators and stopping rules on truncated trajectories, and an adaptive allocation of replicates that concentrates simulation on the candidates near the selection cut-off.

4. **A decision guide and reproducible code** for each screening purpose.

Methods

Data set

We use every trajectory listed by the public HTP-MD web application[4] on 6 to 9 October 2026: 6,270 amorphous polymer/LiTFSI systems simulated with the PCFF+ force field at 353 K, 6,137 for 5 ns and 133 for 50 ns (Table S1). For each trajectory the platform distributes the MSD curves of Li^+ and of the anion at 100 lags, derived transport properties and the raw trajectory, with frames every 2 ps. The storage paths identify each system. Every long run has a 5 ns run of the same system (96 screening polymers and 37 taken from the literature), and, as in the original study,[1] the two runs of such a pair start from the same initial amorphous structure. In addition, 94 screening systems were simulated twice for 5 ns from different initial structures. Every system contains 50 LiTFSI ion pairs at a nominal molality of 1.5 mol kg^{-1}, equilibrated for 5 ns before production;[1] from the starting structures, the systems have 2,328 to 7,541 atoms (median 6,002) in 1 to 15 polymer chains (median 9), in cubic boxes of 3.0 to 4.2 nm side (ion and chain counts verified on 125 complete structure files).

Labels and their time dependence

For an MSD curve averaged over time origins and ions, $\mathrm{MSD}(\tau) = \langle|\mathbf{r}_i(t+\tau) - \mathbf{r}_i(t)|^2\rangle_{i,t}$, we define the apparent diffusivity $D_{\mathrm{app}}(\tau) = \mathrm{MSD}(\tau)/6\tau$. The diffusivity distributed by the platform is the *end-point* estimate $D_{\mathrm{end}}(T) = \langle|\mathbf{r}_i(T) - \mathbf{r}_i(0)|^2\rangle_i/6T$, which uses only the first and last frames;[14] we also use the origin-averaged $D_{\mathrm{app}}(T/2)$. Diffusion corresponds to a lag-independent D_{app} and an MSD exponent $\beta = \mathrm{d}\ln\mathrm{MSD}/\mathrm{d}\ln\tau = 1$. We estimate an effective exponent by a least-squares fit of $\ln\mathrm{MSD}$ against $\ln\tau$ over lags between 20% and 60% of the run, and call a run diffusive when $\beta \geq 0.9$; this is an operational classification, and its

sensitivity to the window is reported in Table S2. For the long runs we compute $D_{\mathrm{app}}(\tau)$ from 0.5 ns to 25 ns and report its ratio to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ and the Spearman correlation of the rankings. The reference is a finite-lag quantity, not a converged coefficient. To test whether the beginning of a run differs from its later part, we split each long run into ten consecutive 5 ns segments and compute, from the data of each segment alone, the end-point and origin-averaged estimates and the effective exponent; each segment is compared with the geometric mean of the segments between 25 and 50 ns.

Noise model, reliability and noise ceiling

All noise statistics are computed on $\ln D$. A label is treated as the property of the system plus the noise of one run, the measurement model also assumed in ref 1. For the replicated systems, the standard deviation of the difference between the two runs divided by $\sqrt{2}$ estimates the noise σ_{run} of one run. With σ_{tot} the spread of all short runs, the reliability of one run is

$$r = 1 - \sigma^2_{\mathrm{run}}/\sigma^2_{\mathrm{tot}}, \tag{1}$$

the fraction of the variance between candidates that reflects differences between systems. We also report the one-way intraclass correlation computed within the replicated subset. A predictor that recovered the property of every system exactly would correlate with one label at most $\sqrt{r}$ and explain at most a fraction r of its variance, the bound used for experimental data[10,11] and known as the noise ceiling in computational neuroscience;[12] the mean of n runs has reliability $nr/[1 + (n-1)r]$.[15,16] To translate reliability into selection we simulate Gaussian populations with the measured variances and record the fraction of the truly best 5%, 10% and 20% recovered when selecting by the mean of n runs.

Two sources of noise are separated as follows. Replicas start from different structures, so their difference contains structural and dynamic noise. The 5 ns run of a pair and the first 5 ns of its long run start from the same structure, so their difference contains only dynamic

noise, σ_{dyn}, and $\sigma^2_{\mathrm{str}} = \sigma^2_{\mathrm{run}} - \sigma^2_{\mathrm{dyn}}$. We test whether the noise depends on mobility by the rank correlation between the absolute difference of two replicas and their mean. Intervals throughout are obtained by bootstrap over systems (2000 resamples).

Selection from paired runs

For the 133 pairs we compare the end-point label of the 5 ns run with the reference of the long run of the same system: median ratio, Spearman correlation (separately for screening and literature polymers) and the recall of the best 5%, 10% and 20% systems, that is, the fraction of the k best according to the reference that are also among the k best according to the short run, against the recall k/n of a random choice. Because the two runs share their initial structure, this comparison contains only the dynamic part of the noise.

Truncated trajectories and stopping rules

From the raw trajectory of each long run we extract the unwrapped lithium positions, subtract the centre-of-mass motion of the whole system as the released library does[14] (validation in the Supporting Information), truncate the trajectory to its first T nanoseconds and estimate the diffusivity from that prefix alone with four estimators: the end-point, the origin-averaged $D_{\mathrm{app}}(T/2)$, one sixth of the slope of a linear fit to the MSD over $0.2T$ to $0.5T$, the conventional procedure,[5] and a power law MSD $= A\tau^{\beta}$ fitted over $0.1T$ to $0.5T$ and extrapolated to τ_{ref}, $\widehat{D_{\mathrm{app}}} = A\tau_{\mathrm{ref}}^{\beta-1}/6$. Estimates are compared with the reference of the full run through the typical multiplicative error $\exp(\mathrm{median}\,|\ln(\hat{D}/D_{\mathrm{ref}})|)$; because the reference shares data with the prefix, we repeat the comparison against references at other lags and against the late slope of the MSD. Stopping rules stop a run when its effective exponent reaches a threshold or when the bootstrap coefficient of variation of an estimate falls below a tolerance; runs that never stop are simulated to 50 ns, and cost and error are averaged over all runs. Thresholds are chosen on a random half of the systems and evaluated on the other half (500 partitions).

Adaptive allocation of replicates

Consider a campaign of $N = 6{,}137$ candidates with a budget of b 5 ns runs per candidate on average, whose goal is the top k. System values are drawn from $\mathcal{N}(0, \sigma^2_{\text{sys}})$ with $\sigma^2_{\text{sys}} = r\sigma^2_{\text{run}}/(1-r)$, and each run adds noise whose standard deviation varies linearly with mobility so as to reproduce the measured noise of the slow and fast halves; every replicate starts from a new structure and carries the full noise. A posterior with a normal prior gives the mean and uncertainty of each candidate. We compare three allocations of the budget. *Uniform*: b runs for every candidate. *Successive halving*:[17] one run for all, then further runs for the best $4k$, $2k$ and k, in rounds that restart from $4k$ until the budget is spent. *Boundary-focused*: one run for all, then batches of 5% of N further runs to the candidates with the largest $p(1-p)$, where p is the posterior probability of lying above the current top-k cut-off, in the spirit of optimal computing budget allocation.[18] The final selection is the top k by posterior mean, and the metric is the recall of the true top k, averaged over 200 simulated campaigns. Robustness is checked with the empirical shape of the short-run distribution instead of a normal one, with the origin-averaged label, and with a constant instead of a mobility-dependent noise.

The public data allow a partial check on real runs. In the replicated systems, the first run is the original screening run and the second starts from a new structure, and 92 of them also have a long run, which starts from the same structure as the first. We use them in two ways, with noise parameters estimated leaving out the system being evaluated. First, from one run of each system the model predicts the probability that the other run exceeds the top-k cut-off; we compare these predictions with what happens (Brier score against the base rate and calibration slope). Second, we run a small real campaign: every system receives its first run, and 46 second runs go either to the candidates with the largest $p(1-p)$ or to randomly chosen ones; the selection is evaluated against the long runs, which are independent of the second runs.

Results and Discussion

Labels depend on the simulated time

After 5 ns the median effective exponent of the lithium MSD is 0.43 (10th to 90th percentile range 0.26 to 0.60): 97.8% of the runs have $\beta < 0.7$ and 2 of 6,137 meet the diffusive criterion (Fig. 1). Ten times more simulation moves the distribution but does not resolve the problem: at 50 ns the median is 0.65 and 5 of 133 runs are diffusive (0 to 19 depending on the fitting window, Table S2). In the long runs the apparent diffusivity falls steadily with the lag (Fig. 2): at 5 ns it is a median 1.95 times its value at 25 ns (10th to 90th percentile range 1.47 to 2.40), and only 73.7% of the systems are within 25% of the reference at 15 ns (Table S4). Rankings are far more stable: the Spearman correlation with the ranking at 25 ns is 0.91 already at 0.5 ns and 0.96 at 5 ns. A short-run label is therefore an apparent, window-dependent quantity whose value should not be compared across studies with different lengths, but whose ordering carries information.

Labels also depend on *when* the time is simulated (Fig. 3). Analysed with its own data only, the first 5 ns segment of a long run is a median 1.63 times faster than the segments between 25 and 50 ns (95% interval 1.52 to 1.81; 1.47 with the origin-averaged estimator), and it is faster in 93% of the systems (Wilcoxon signed-rank $p = 3.6 \times 10^{-22}$). The slowdown is not a short transient: the apparent diffusivity keeps falling by a factor of 0.92 every 10 ns between 5 and 50 ns, in 85% of the systems, and the effective exponent of the segments decreases from 0.48 in the first to 0.31 in the late ones. Lithium mobility therefore keeps decreasing throughout the runs, despite the 5 ns equilibration that precedes production;[1] we measure this slowdown, not its cause, which is consistent with a slow conformational relaxation of the polymer or of the lithium coordination network that the equilibration does not complete. The effect is stronger for the screening polymers (1.80) than for those taken from the literature (1.28).

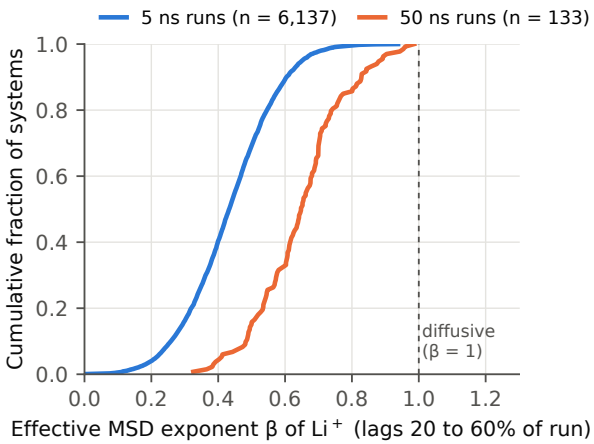


Figure 1: Cumulative distribution of the effective MSD exponent β of Li^+ in the 5 ns and 50 ns runs. The dashed line marks diffusive motion.

Run-to-run noise and its sources

Two 5 ns runs of the same system from different structures differ by a typical factor of 1.43 for the end-point label ($\sigma_{\mathrm{run}} = 0.35$ in log units) and 1.32 for the origin-averaged one (Table S5). Against the spread of the whole screening set, the reliability of one end-point label is 0.47 (95% interval 0.31 to 0.63), and 0.58 (0.45 to 0.70) for the origin-averaged label; the intraclass correlation within the replicated subset agrees (0.53 and 0.62). Roughly half of the variance between screened candidates is therefore noise.

The noise has two comparable sources. The 5 ns run of each pair agrees with the first 5 ns of its long run, which starts from the same structure, to a median ratio of 1.00 with a noise of 0.24, against 0.35 between replicas from different structures: the initial structure accounts for 55% (31 to 72%) of the run-to-run variance and the dynamics for the rest. Re-sampling the structure and averaging the dynamics both reduce the noise. A replicate that starts from a new structure samples both sources, as ensembles of independent runs do in other MD applications,[19] but it also costs a new equilibration; whether it beats extending the same run at equal cost cannot be decided with these data. This does not affect the allocation results below, in which every run, uniform or adaptive, starts from a new structure. The noise also depends on mobility: it is 0.40 in the slower half of the replicated systems and

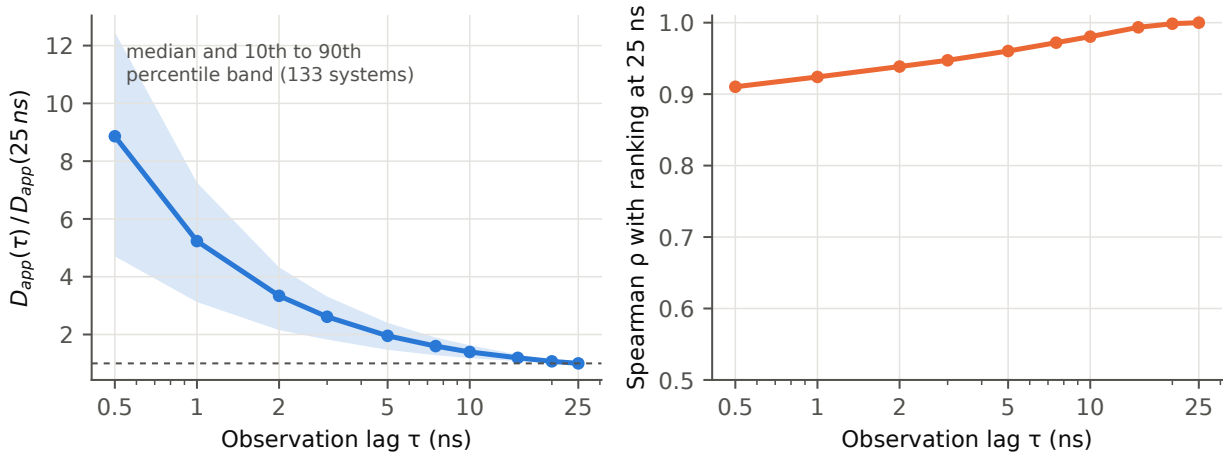


Figure 2: Lag dependence in the 133 runs of 50 ns. Left: apparent diffusivity of Li^+ relative to its value at 25 ns (median and 10th to 90th percentile band). Right: Spearman correlation between the rankings at lag τ and at 25 ns.

0.30 in the faster half (rank correlation between the absolute difference and the level −0.26, $p = 0.01$), whereas the learning model of the original study assumes noise independent of the polymer.[1] Two candidates measured once each differ by a factor of 1.65 from noise alone at one standard deviation and up to 2.67 at the 95% level.

A noise ceiling for learned predictors

The reliability bounds what any predictor can achieve against short-run labels (Table 1). A model that recovered the property of every system exactly would correlate with a single end-point label at most 0.69 (95% interval 0.55 to 0.79) and explain at most $R^2 = 0.47$ of its variance; with the origin-averaged label the ceilings are 0.76 and 0.58. Choosing the best 10% by one end-point label recovers 45% of the truly best tenth, which is also the apparent recall of a perfect model evaluated against such labels. Averaging runs raises the ceiling: two end-point runs give a reliability of 0.64, four give 0.78, and a reliability of 0.8 requires 5 runs (3 with the origin-averaged label). The original study estimated that its graph network predicts the conductivity with the accuracy of about four simulations per polymer;[1] for the lithium diffusivity studied here, four runs correspond to a reliability of 0.78.

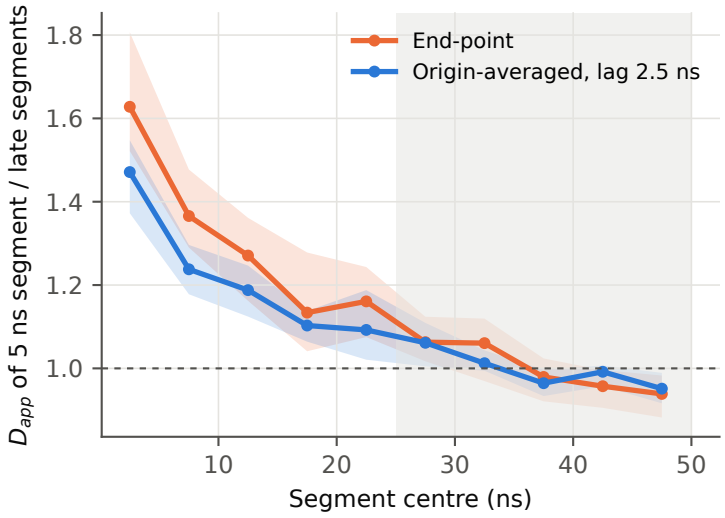


Figure 3: Apparent diffusivity of consecutive 5 ns segments of the 133 long runs, each computed from its own data, relative to the segments between 25 and 50 ns (shaded): median over systems and 95% bootstrap interval.

A real model reaches this ceiling (Table 2). We trained a random forest on Morgan count fingerprints (radius 2, 2048 bits) of the monomer of each of the 6,042 screened systems, with the end-point label of its original 5 ns run as target, and obtained out-of-sample predictions by 5-fold cross-validation split by system. The predictions correlate with the labels with Pearson $r = 0.71$ (95% interval 0.70 to 0.72; $R^2 = 0.50$), statistically indistinguishable from the estimated ceiling of 0.69 (95% interval 0.55 to 0.79). That the point estimate exceeds the ceiling slightly suggests that the reliability of the whole population is somewhat higher than the replicated subset indicates; either way, a simple model is already close to what single short-run labels allow, so a better architecture could hardly show an advantage on them. The replicated systems are consistent with the measurement model. For the same predictions, the correlation rises from 0.67 and 0.69 against either single label to 0.78 against the mean of the two, as the reliability of a two-run mean predicts (0.79; gain 95% interval 0.07 to 0.13). On such labels, better labels rather than better models are what can raise the measured performance.

Against the long runs, the model, which never saw these systems, ranks the 50 ns reference as well as the system's own 5 ns run among the screening polymers (Spearman 0.67 against

0.68), but worse among the literature polymers (0.67 against 0.93), whose chemistry lies outside the screened space. Because each long run shares its initial structure with the short run, this comparison favours the short run; a model trained on short labels is thus a substitute for a short simulation within the chemical space it was trained on, not beyond it.

Two practical consequences follow. A test metric computed against single short-run labels cannot exceed the ceiling, so a model close to it is as good as the labels allow, and progress has to be measured against averaged labels. And because 94 systems appear twice in the public dataset, splits should be made by system; in this dataset the effect of splitting by trajectory instead is small (0.66 against 0.68 on the 168 replicated trajectories), because the model agrees with either run of a replicated system more closely than the two runs agree with each other (rank correlation 0.48), so the partner's label adds little.

Table 1: Noise ceiling of short-run labels in the screening population: reliability of the mean of n runs, maximum correlation of a perfect predictor with that label, and recall of the truly best systems when selecting by it.

				Recall of the true top		
Label	Runs averaged	Reliability	Max. correlation	5%	10%	20%
End-point	1	0.47	0.69	38%	45%	56%
	2	0.64	0.80	49%	56%	64%
	3	0.73	0.85	56%	63%	70%
	4	0.78	0.88	61%	66%	73%
Origin-averaged, lag $T/2$	1	0.58	0.76	45%	52%	61%
	2	0.73	0.85	56%	63%	70%
	3	0.80	0.90	63%	68%	75%
	4	0.84	0.92	67%	72%	77%

Table 2: Out-of-sample correlation of a random forest on monomer fingerprints with short-run labels (cross-validation split by system), against the noise ceiling of each label.

Target of the out-of-sample prediction	Systems	Pearson r (95% CI)	Ceiling (95% C
One 5 ns label, all systems	6,042	0.71 (0.70 to 0.72)	0.69 (0.55 to 0.7
One 5 ns label (first / second run), replicated systems	94	0.67 / 0.69	0.69 (0.55 to 0.7
Mean of two 5 ns labels, replicated systems	94	0.78 (0.66 to 0.86)	0.80 (0.68 to 0.8

Selecting candidates from one short run

For the 133 systems simulated at both lengths, the end-point label of the 5 ns run is a median 2.90 times the long-run reference (10th to 90th percentile range 1.58 to 4.74; Fig. S2), and the Spearman correlation between them is 0.84 (95% interval 0.76 to 0.90). The correlation owes much to the wide range of mobilities in this subset: it is 0.93 among the 37 literature polymers but 0.68 among the 96 screening polymers. In the whole paired set the short run identifies 11 of the 13 best systems (Table 3), but these top sets consist almost entirely of literature polymers (13 of 13 and 24 of 27), which are easy to separate. Within the 96 screening polymers the short run still recovers 7 of the 10 best and 14 of the 19 best, against 10% and 20% by chance. These recalls are optimistic, because the short and long runs of a pair share their initial structure and only the dynamic part of the noise separates them. A short run thus separates good from poor electrolytes but cannot resolve similar ones, in line with the noise ceiling above.

The median ratio of 2.90 is larger than the 1.95 obtained at the same lag from the origin-averaged long curves, and this slowdown explains the gap. The short runs behave like the first 5 ns of the long runs (median ratio 1.00), and that first segment is 1.42 times faster than the average of all segments, which is roughly what separates a single early window from a curve averaged over all time origins. Every short label in the dataset is therefore biased upwards by the early, faster part of the trajectory, and more so for the screening polymers. For ranking, this bias matters only to the extent that it differs between systems.

Spending the budget within a run: estimators and stopping

When only the first T nanoseconds of a long run are available (Fig. 4, Table S6), the end-point, half-lag and linear-fit estimates overestimate the reference at every short length, by typical factors of 2.83, 4.05 and 2.02 at 5 ns. Their bootstrap intervals seldom contain the reference (3%, 0% and 39% of the runs): the intervals describe the apparent diffusivity at the lags available in the prefix, so a within-run error bar says nothing about the distance

Table 3: Recall of the best systems according to the 50 ns reference when they are selected from the 5 ns run of the same system (133 systems; the last rows only the 96 screening polymers). Intervals by bootstrap over systems; "Random" is the expected recall of a random selection of k systems. All recalls are optimistic, because the two runs of a pair share their initial structure.

Estimator, 5 ns run	Top	k	Hits	Recall (95% CI)	Random
End-point	5%	7	7	100% (50 to 100)	5%
	10%	13	11	85% (62 to 100)	10%
	20%	27	23	85% (71 to 94)	20%
Origin-averaged, lag $T/2$	5%	7	6	86% (50 to 100)	5%
	10%	13	10	77% (62 to 100)	10%
	20%	27	22	81% (75 to 94)	20%
End-point, screening only	5%	5	4	80% (67 to 100)	5%
	10%	10	7	70% (50 to 86)	10%
	20%	19	14	74% (54 to 83)	20%

to the long-lag value. A power-law extrapolation removes most of the systematic deviation, with a median ratio of 0.95 and a typical error of 1.37 from 5 ns. It keeps a typical error of 1.33 to 1.45 against references at other lags, but underestimates the late slope of the MSD (median ratio 0.54; Table S7): lithium is still accelerating towards diffusion, so the power law predicts the apparent diffusivity within the observed window, not the long-time coefficient.

On a single cost-error plane (Fig. 5), fixed-length power-law extrapolation forms the lower envelope. A typical error of $\times 1.5$ requires 2.0 ns with it, against 12.0 ns with a linear fit and 20.0 ns with the end-point label; for $\times 1.3$ the figures are 10.0, 20.0 and 30.0 ns. Adaptive stopping rules save nothing: the regime rules rarely fire, and precision rules stop early only at the price of large errors (Table S8). A setting chosen exactly at the boundary of a target often misses it on held-out systems (Table S9), so a margin of one step is advisable.

Spending the budget across candidates: adaptive allocation

The noise model measured above is compatible with the replicated systems: the differences between replicas show no departure from a Gaussian (excess kurtosis 0.13, Shapiro-Wilk $p = 0.81$). With it, how replicates are distributed matters more than how many are run

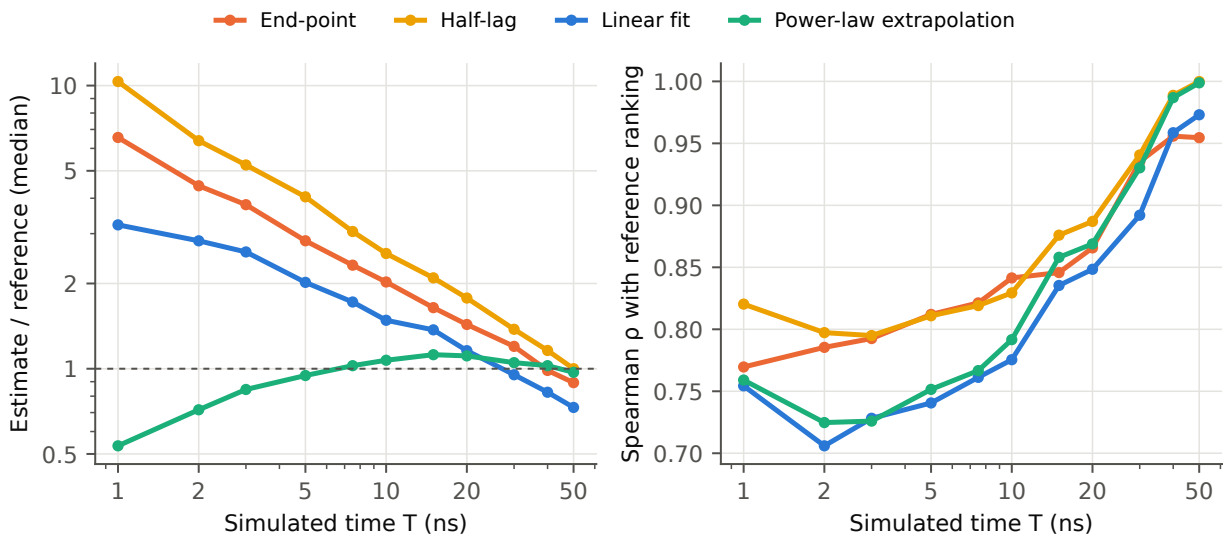


Figure 4: Estimators computed from the first T nanoseconds of the 133 long runs. Left: median ratio to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ of the full run. Right: Spearman correlation with the reference ranking.

(Fig. 6, Table 4). Uniform replicates improve slowly: for the top 10% of 6,137 candidates the recall rises from 48% with one run to 72% with four. Boundary-focused allocation reaches 75% with two runs per candidate on average and 84% with four, and matches four uniform replicates with 1.8 runs per candidate, a saving of $\times 2.2$. The saving grows with the selectivity of the target: $\times 2.7$ for the top 5% and $\times 3.2$ for the top 1% (Table S10). Successive halving, which re-runs the current leaders in repeated rounds, performs similarly: with the end-point label it is better for the top 1% at small budgets (74% against 68% with 1.25 runs per candidate), and with the origin-averaged label it is worse at larger budgets (84% against 89% with four). The gain therefore comes from adaptivity, not from a particular rule. The result holds with the empirical distribution of the short runs and with the origin-averaged label, and knowing that slow systems are noisier does not improve the allocation (Supporting Information). In practice, a campaign should give one short run to every candidate and spend the rest of the budget on new-structure replicates of the candidates whose membership of the top set is uncertain.

These gains are predictions of the noise model, since the public data contain at most two

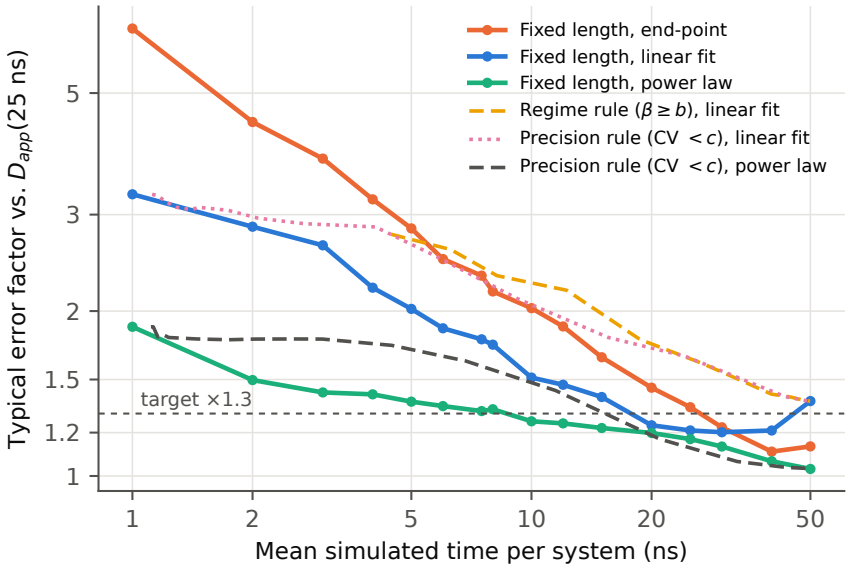


Figure 5: Cost-error frontier of estimators and stopping rules on the 133 long runs. Each line follows one rule as its parameter varies; runs that do not stop before 50 ns count as 50 ns.

runs per system; the replicated systems allow only a partial check, which is consistent with them but has little statistical power (Table S11). The model's probabilities of exceeding the top-10% cut-off are informative on real runs, improving the Brier score of the base rate by 36% with a calibration slope of 1.10 (95% interval 0.65 to 2.28). In the real campaign on 92 systems, sending the 46 second runs to the boundary recovers 78% of the top 10% according to the long runs, as many as giving a second run to every system (78%), against 71% for randomly placed second runs and 67% with first runs only. With 9 systems in the top set the difference from random placement is not significant (95% interval from −17 to +17 percentage points), and for the top 20% no strategy improves on the first run alone; the long runs share their structure with the first runs, which makes this check conservative for any second run from a new structure.

A second property: the anion diffusivity

To test whether these conclusions depend on the property, we repeated the main analyses for the diffusivity of the TFSI anion (Table 5). Because the platform distributes only the

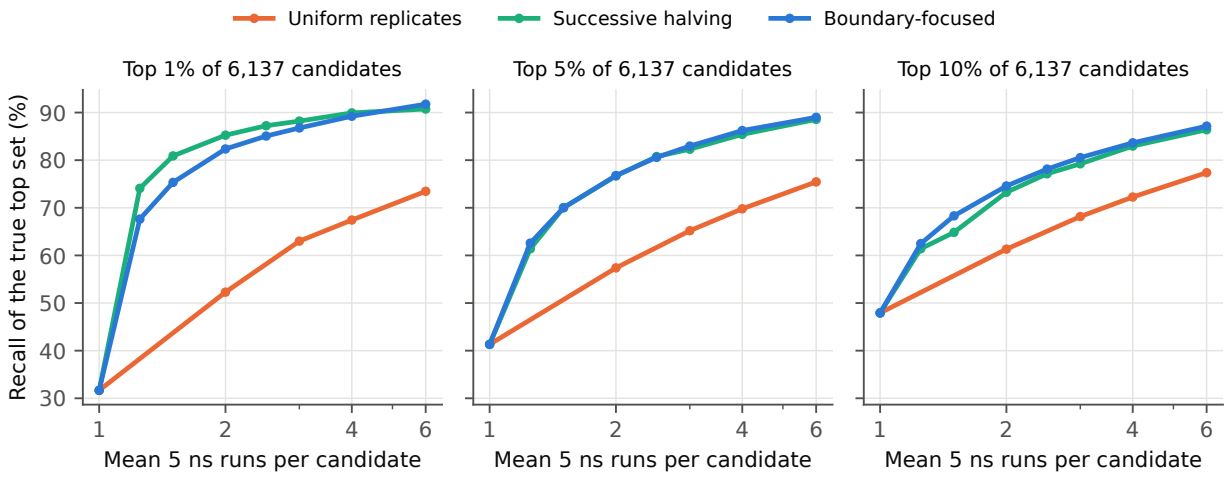


Figure 6: Recall of the true top set as a function of the mean number of 5 ns runs per candidate, for three ways of allocating the budget (200 simulated campaigns of 6,137 candidates with the measured noise).

origin-averaged estimate for the anion, both ions are compared with that estimator. The anion label is much less noisy: its reliability is 0.80 (95% interval 0.75 to 0.85) against 0.58 for lithium, so its noise ceiling is higher (0.89 against 0.76) and a short run ranks the screening polymers better against the long runs (Spearman 0.83 against 0.70). The structure of the results, however, is the same. The random forest reaches the ceiling for both ions (0.86 and 0.77), its correlation with the mean of two replicas rises as the measurement model predicts (0.90 against an expected 0.90 for the anion), and adaptive allocation matches four uniform replicates with less than half the runs ($\times$2.7 and $\times$2.3 for the top 10%). The numbers are specific to each property, but for these two properties the conclusions are the same.

Practical guidance

Table 6 turns these results into recommendations, ordered from the least to the most demanding purpose. They apply to the setting studied (LiTFSI in amorphous polymers, PCFF+, 353 K); the lengths are those for which this study provides evidence, not universal minima. Three recommendations hold for every purpose: report the lag and the estimator of any transport label together with the effective exponent of the population, prefer origin-

Table 4: Recall of the true top set for each allocation and mean budget (end-point label).

		Mean runs per candidate					
Target	Allocation	1	1.5	2	3	4	6
Top 1%	Uniform replicates	32%	n/a	52%	63%	67%	73%
	Successive halving	32%	81%	85%	88%	90%	91%
	Boundary-focused	32%	75%	82%	87%	89%	92%
Top 5%	Uniform replicates	41%	n/a	57%	65%	70%	75%
	Successive halving	41%	70%	77%	82%	85%	89%
	Boundary-focused	41%	70%	77%	83%	86%	89%
Top 10%	Uniform replicates	48%	n/a	61%	68%	72%	77%
	Successive halving	48%	65%	73%	79%	83%	86%
	Boundary-focused	48%	68%	75%	81%	84%	87%

Table 5: Main results for the lithium and the TFSI anion diffusivities, both with the origin-averaged estimator at half the run length. Noise statistics from the 94 replicated systems; random forest and allocation as in the previous sections (allocation with constant noise).

	Li^+	$TFSI^-$
Run-to-run noise (log)	0.28	0.19
Noise, slow / fast half	0.29 / 0.26	0.20 / 0.18
Reliability (95% CI)	0.58 (0.44 to 0.70)	0.80 (0.75 to 0.85)
Noise ceiling	0.76	0.89
Random forest vs one label	0.77	0.86
Random forest vs two-run mean (expected)	0.84 (0.86)	0.90 (0.90)
Spearman with 50 ns run, screening	0.70	0.83
Spearman with 50 ns run, literature	0.94	0.96
Runs matching 4 uniform replicates (top 10%)	1.7 (×2.3)	1.5 (×2.7)

averaged estimates to end-point ones, and never read a within-run error bar as the distance to the long-time value.

Limitations

All systems share one force field, one temperature and one salt; the conclusions hold for two properties, the lithium and anion diffusivities, but other properties, such as the conductivity, may have different noise. The long runs were chosen by the dataset's authors rather than at random. The reference lag of 25 ns is not a converged limit and shares data with the

Table 6: Decision guide for MD screening, from the results of this study.

Purpose	What to do	Evidence in this study
Separate good from poor candidates	One 5 ns run each; rank by origin-averaged D_{app}	Spearman 0.94 among widely different polymers
Find the best candidates under a fixed budget	One run each, then new-structure replicates for candidates near the cut-off	In simulations with the measured noise, matches four uniform replicates with 1.8 runs per candidate for the top 10% and 1.2 for the top 1%; real runs consistent but underpowered
Decide between two similar candidates	Several new-structure runs each	One run each: noise alone gives differences of $\times$1.65 (1 SD) and up to $\times$2.67 (95%)
Train or evaluate a model on screening labels	Split by system; average replicates where available; compare test metrics with the ceiling	Correlation with single labels capped near 0.69 (95% CI 0.55 to 0.79); a fingerprint random forest reaches 0.71, and 0.78 against the mean of two runs
Estimate the apparent diffusivity at a stated lag	5 to 10 ns with power-law extrapolation, one step above the cheapest setting	Typical error 1.37 from 5 ns against 2.02 for a linear fit; underestimates the late MSD slope
Obtain a diffusion coefficient or compare with experiment	Simulate until the exponent is compatible with one and D_{app} is flat across lag windows	5 of 133 runs of 50 ns meet the criterion

truncated trajectories, so prefix-based errors at long T are optimistic. Self-diffusion is a proxy for conductivity, which also depends on ion correlations,[20] and finite-size effects[21] are not corrected. The noise model rests on 94 replicated pairs: the allocation gains are predictions of a model that the real runs support but, with 92 systems, cannot confirm with statistical significance; a campaign with several replicates per candidate would test them directly.

Conclusions

Short MD runs make useful but imperfect labels. In the largest public set of polymer electrolyte simulations, a 5 ns label measures an apparent, time-dependent quantity that is inflated by a slow decrease of lithium mobility during the run, and about half of its variance across candidates is run-to-run noise that comes in comparable parts from the initial structure

and from the dynamics. This caps what any model can learn from such labels, and a simple fingerprint model is already close to that ceiling; it also limits a single run to separating good from poor candidates. The same noise model shows how to spend a budget: in simulations with the measured noise, one run per candidate and further runs only where the selection is uncertain find the best candidates as reliably as four uniform replicates at half the cost or less, a prediction that real runs support but cannot yet confirm, whereas stopping individual runs adaptively saves nothing. The same picture holds for the anion diffusivity, whose labels are less noisy. Transport coefficients in the long-time limit remain out of reach for most systems even with tenfold longer runs.

Data and Software Availability

The HTP-MD trajectories and curves are publicly available from the Toyota Research Institute platform.[4] The analysis scripts, the per-trajectory result tables and the scripts that generate every number, table and figure of this paper are available at `https://github.com/fhernandogallego/md-label-reliability` and archived on Zenodo (`https://doi.org/10.5281/zenodo.23483460`); the raw trajectories are not redistributed.

Supporting Information Available

Data set details and validation of the trajectory processing; sensitivity of the regime census and of the lag dependence; run-to-run reproducibility; end-point estimator; paired runs; prefix-based estimators, stopping rules and held-out evaluation; robustness of the adaptive allocation and its check on real runs (PDF).

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Notes

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